

Rainfall Data Analysis Assignment (100 Marks, Due 6th July 2025)

Group Members

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You are provided with a rainfall dataset recording monthly measurements from various weather stations across different regions. The dataset includes rainfall (mm), number of rainy days, maximum and minimum temperatures, and whether the station is in an urban or rural area.

Dataset columns:

- **Region**: Name of the region (e.g., North, South, East, West)
- **Year**: Year of measurement
- **Month**: Month (e.g., Jan, Feb, ..., Dec)
- **Station**: Weather station name/ID (e.g., StationA, StationB)
- **Rainfall_mm**: Rainfall amount in millimeters
- **RainyDays**: Number of days in the month with rain
- **MaxTemp_C**: Maximum temperature recorded in that month (°C)
- **MinTemp_C**: Minimum temperature recorded in that month (°C)
- **UrbanRural**: "Urban" or "Rural"

Part 1: Data Exploration (15 Marks)

1. Load the data into a pandas DataFrame. (2 Marks)
2. Display the first 5 rows of the DataFrame. (2 Marks)
3. Check for and handle any missing values. (6 Marks)
4. Provide summary statistics for all numeric columns. (5 Marks)

```
import pandas as pd
# question 1
rainfall_data = pd.read_csv("Rainfall.csv")

# question 2
print (rainfall_data.head(5))
print (" ")

# question 3
missing_values = rainfall_data.isnull().sum() # total up the column
print (missing_values)
print (missing_values[missing_values>0])
missing_values.dropna(inplace=True)
```

```
# question 4
```

```
rainfall_data.describe()
```

```
   Region  Year  Month  Station  Rainfall_mm  RainyDays  MaxTemp_C
MinTemp_C \
0  North  2023   Jan  StationA         56.2         9         12.1
2.5
1  North  2023   Feb  StationA         72.1        12         14.7
3.0
2  North  2023   Jan  StationB         60.5        10         11.2
1.8
3  South  2023   Jan  StationC         81.0        13         19.6
7.2
4  South  2023   Feb  StationC         63.4        11         20.1
7.7
```

```
   UrbanRural
0      Urban
1      Urban
2      Rural
3      Urban
4      Urban
```

```
Region      0
Year        0
Month        0
Station      0
Rainfall_mm  0
RainyDays    0
MaxTemp_C    0
MinTemp_C    0
UrbanRural   0
dtype: int64
Series([], dtype: int64)
```

```
   Year  Rainfall_mm  RainyDays  MaxTemp_C  MinTemp_C
count  10.0    10.000000    10.00000    10.000000    10.000000
mean   2023.0    66.160000    10.90000    16.100000     4.740000
std      0.0    12.154121     1.66333     2.982542     2.029888
min    2023.0    45.700000     8.00000    11.200000     1.800000
25%    2023.0    59.825000    10.00000    14.775000     3.050000
50%    2023.0    62.450000    11.00000    16.100000     5.100000
75%    2023.0    76.075000    12.00000    18.125000     6.050000
max    2023.0    84.200000    13.00000    20.100000     7.700000
```

Part 2: Grouping and Aggregation (30 Marks)

1. For each **Station**, compute: (6 Marks)
 - Total rainfall

- Average number of rainy days
- Average maximum temperature
- Average minimum temperature

```
stations = rainfall_data.groupby('Station').agg(
    Total_Rainfall = ('Rainfall_mm', 'sum'),
    Average_Days = ('RainyDays', 'mean'),
    Average_Max_Temp = ('MaxTemp_C', 'mean'),
    Average_Min_Temp = ('MinTemp_C', 'mean')
)
```

```
print(stations)
```

	Total_Rainfall	Average_Days	Average_Max_Temp
Average_Min_Temp Station			
StationA	128.3	10.5	13.40
2.75			
StationB	122.0	10.5	13.10
2.50			
StationC	144.4	12.0	19.85
7.45			
StationD	105.3	9.0	16.65
5.90			
StationE	161.6	12.5	17.50
5.10			

1. For each **Region & UrbanRural** combination, calculate: (6 Marks)

- Average rainfall
- Average maximum temperature
- Average number of rainy days

```
Regions = rainfall_data.groupby(['Region', 'UrbanRural']).agg(
    Avg_Rainfall = ('Rainfall_mm', 'sum'),
    Avg_Max_Temp = ('MaxTemp_C', 'mean'),
    Avg_Days = ('RainyDays', 'mean')
)
```

```
print(Regions)
```

		Avg_Rainfall	Avg_Max_Temp	Avg_Days
Region	UrbanRural			
East	Rural	105.3	16.65	9.0
North	Rural	122.0	13.10	10.5
	Urban	128.3	13.40	10.5
South	Urban	144.4	19.85	12.0
West	Urban	161.6	17.50	12.5

1. For each **Month** and **Region**, calculate: (6 Marks)

- Total rainfall
- Mean rainfall
- Minimum and maximum rainfall

```
MonthsR = rainfall_data.groupby(['Month', 'Region']).agg(
    {'Rainfall_mm': ['sum', 'mean', 'min', 'max']}
)

print(MonthsR)
```

		Rainfall_mm			
		sum	mean	min	max
Month	Region				
Feb	East	59.6	59.60	59.6	59.6
	North	72.1	72.10	72.1	72.1
	South	63.4	63.40	63.4	63.4
	West	84.2	84.20	84.2	84.2
Jan	East	45.7	45.70	45.7	45.7
	North	116.7	58.35	56.2	60.5
	South	81.0	81.00	81.0	81.0
	West	77.4	77.40	77.4	77.4
Mar	North	61.5	61.50	61.5	61.5

1. Find the **Region and Month** with the highest recorded rainfall in the dataset. (4 Marks)

```
highest_rainfall = (
    rainfall_data.groupby(['Region', 'Month'])
    ['Rainfall_mm'].max()
    .sort_values(ascending=False)
    .head(1)
)

print(highest_rainfall)
```

Region Month
West Feb 84.2
Name: Rainfall_mm, dtype: float64

1. For each **Year** and **UrbanRural** group, calculate: (4 Marks)
 - Average rainfall
 - Minimum rainfall
 - Maximum rainfall

```
YearUR = rainfall_data.groupby(['Year', 'UrbanRural']).agg(
    {'Rainfall_mm': ['mean', 'min', 'max']}
)

print(YearUR)
```

		Rainfall_mm		
		mean	min	max

Year	Urban	Rural		
2023	Rural	56.825000	45.7	61.5
	Urban	72.383333	56.2	84.2

1. For each **Region**, calculate the average temperature range per month (i.e., average of MaxTemp_C - MinTemp_C). (4 Marks)

```
rainfall_data['Temp_range'] = rainfall_data['MaxTemp_C'] -
rainfall_data['MinTemp_C']
```

```
Regions_Avg = rainfall_data.groupby('Region').agg(
    Avg_MinRain = ('MinTemp_C', 'mean'),
    Avg_MaxRain = ('MaxTemp_C', 'mean'),
    Range = ('Temp_range', 'mean')
)
```

```
print(Regions_Avg)
```

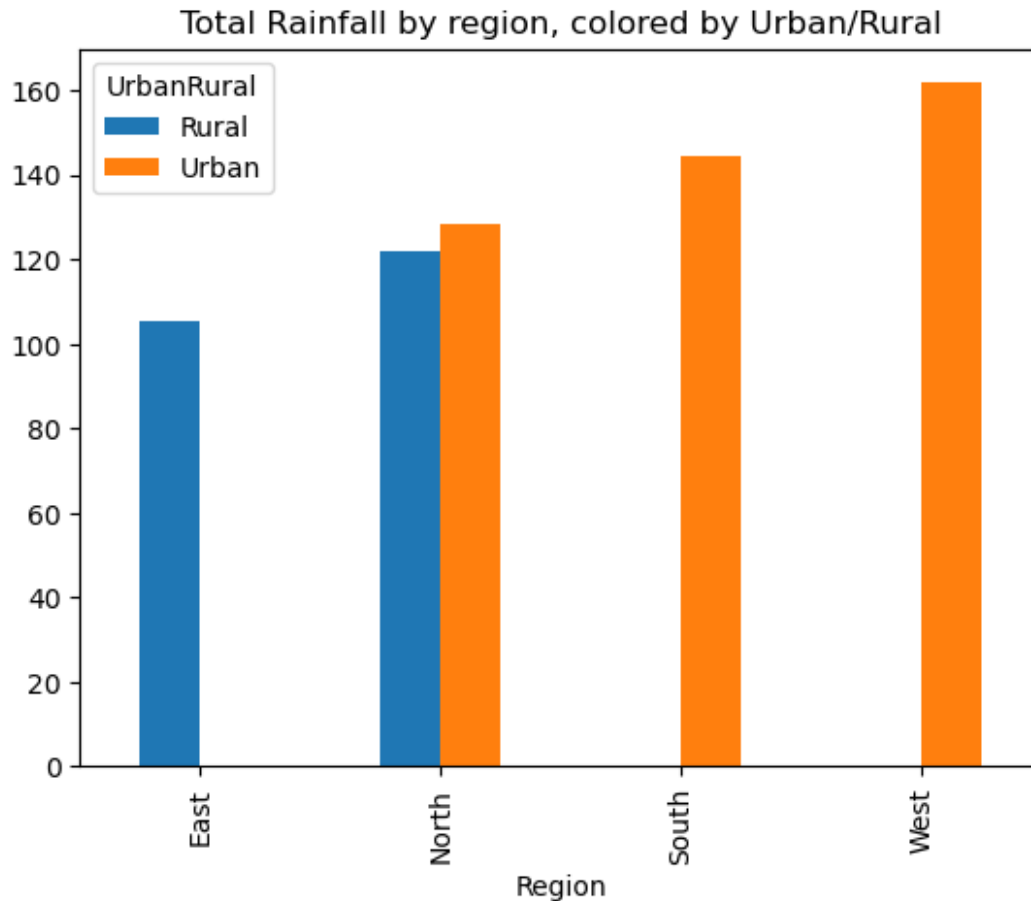
	Avg_MinRain	Avg_MaxRain	Range
Region			
East	5.900	16.65	10.750
North	2.625	13.25	10.625
South	7.450	19.85	12.400
West	5.100	17.50	12.400

Part 3: Data Visualization (matplotlib) (40 Marks)

1. **Bar Chart:** (8 Marks)
 - Plot total rainfall by region, colored by Urban/Rural.
 - Add value labels, gridlines, axis labels, title, and adjust bar width for clarity.

```
import matplotlib.pyplot as plt

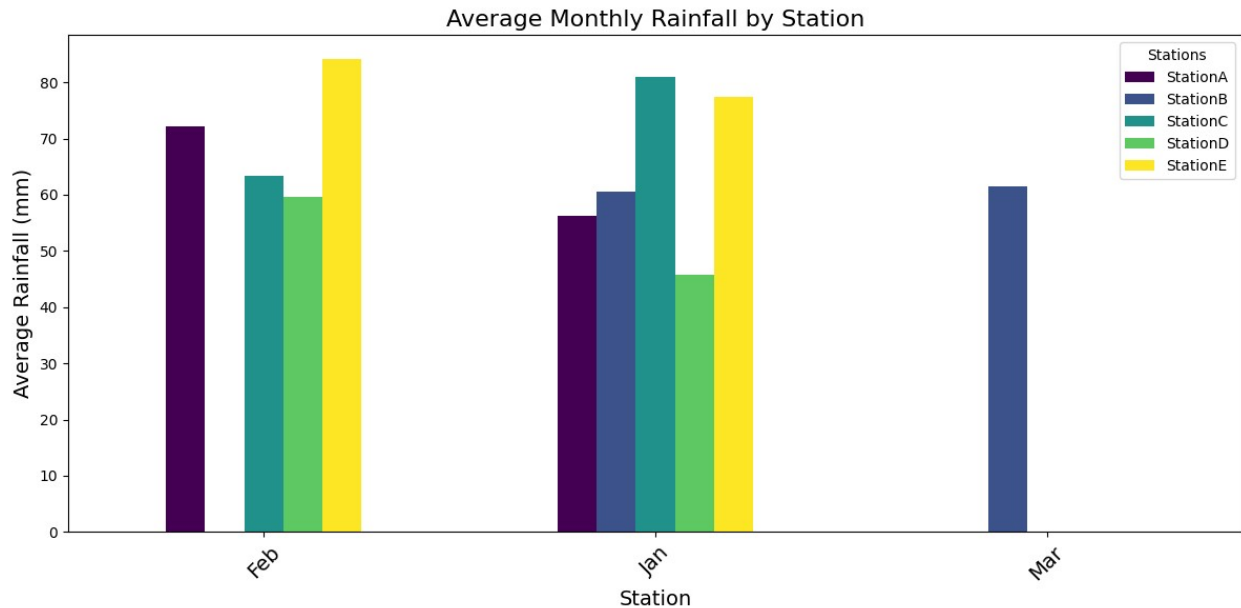
total_rain = rainfall_data.groupby(['Region', 'UrbanRural'])
['Rainfall_mm'].sum().unstack()
%matplotlib inline
total_rain.plot.bar()
plt.title('Total Rainfall by region, colored by Urban/Rural')
# plt.show()
```



1. **Grouped Bar Chart:** (6 Marks)

- Show average monthly rainfall for each station.
- Use different colors for each station, include a legend, rotate x-tick labels, and increase figure size for readability.

```
average = rainfall_data.groupby(['Month', 'Station'])
['Rainfall_mm'].mean().unstack()
fig, ax = plt.subplots(figsize=(12, 6))
average.plot(kind='bar', ax=ax, colormap='viridis')
ax.set_xticklabels(average.index, rotation=45,
ha='center', fontsize=14)
ax.set_title('Average Monthly Rainfall by Station', fontsize=16)
ax.set_ylabel('Average Rainfall (mm)', fontsize=14)
ax.set_xlabel('Station', fontsize=14)
ax.legend(title='Stations') # can customize legend
plt.tight_layout()
```



1. Line Plot: (6 Marks)

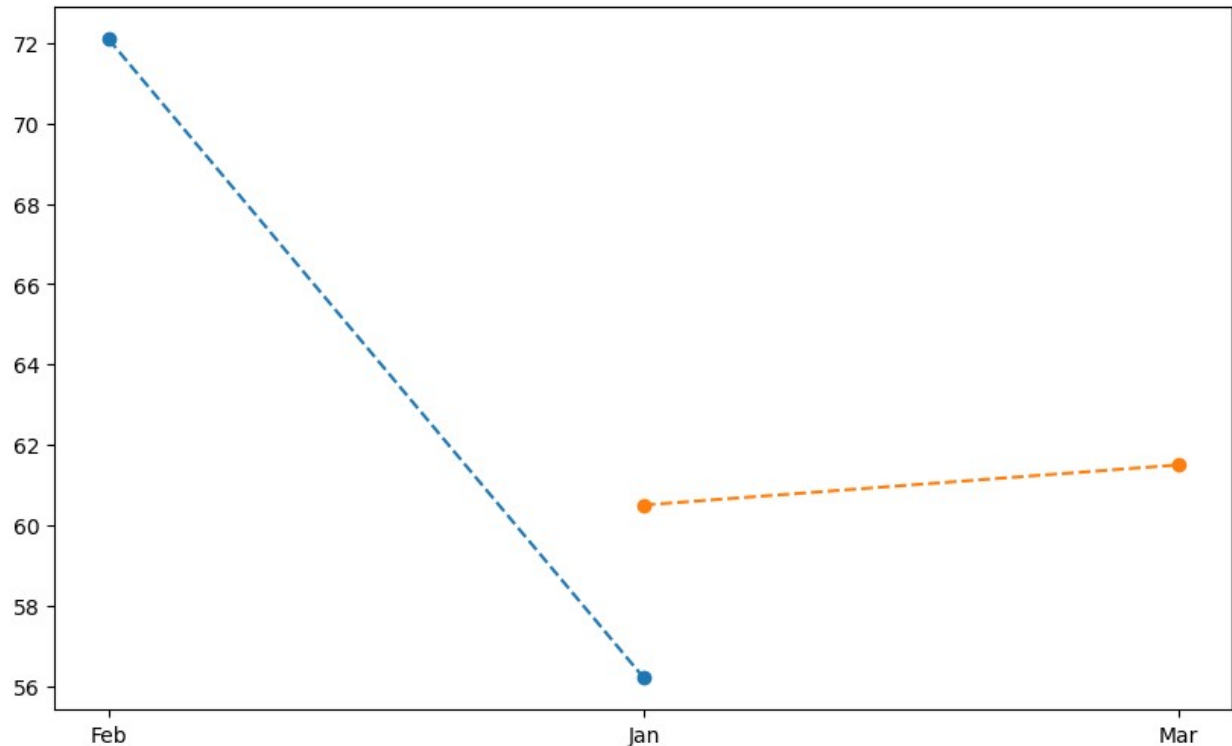
- Show the trend of rainfall over months for at least two selected stations.
- Use markers, dashed lines, highlight the month with maximum rainfall, and format the x-axis for months.

```
stations = rainfall_data['Station'].unique()[1:2]
subset = rainfall_data[rainfall_data['Station'].isin(stations)]
monthly_trend = (
    subset.groupby(['Month', 'Station'], observed=True)['Rainfall_mm']
    .mean()
    .reset_index()
)

pivot = monthly_trend.pivot(index='Month', columns='Station',
                             values='Rainfall_mm')
fig3, ax3 = plt.subplots(figsize=(10,6))

for station in pivot.columns:
    ax3.plot(
        pivot.index,
        pivot[station],
        marker='o',
        linestyle='--',
        label=station
    )

# plt.show()
```



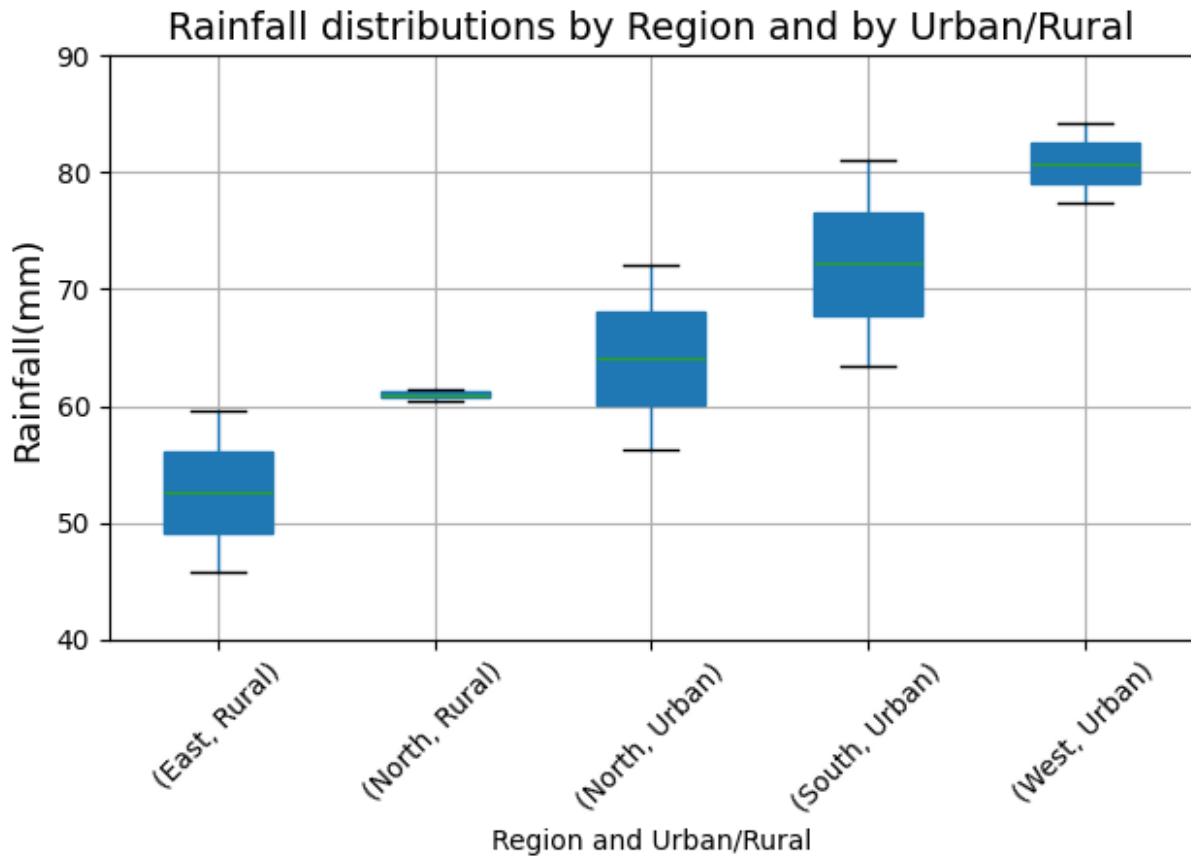
1. **Boxplot:** (6 Marks)

- Display rainfall distributions by region and by Urban/Rural.
- Use custom box colors, highlight outliers, and set y-axis limits.

```

boxplot = rainfall_data.boxplot(column='Rainfall_mm', by=['Region',
'UrbanRural'],
                                patch_artist=True,
                                flierprops=dict(marker='o',
markerfacecolor='red', markersize=8),
                                )
plt.ylim(40,90)
plt.suptitle('')
plt.title('Rainfall distributions by Region and by Urban/Rural',
fontSize=14)
boxplot.set_ylabel('Rainfall(mm)', fontsize=13)
boxplot.set_xlabel('Region and Urban/Rural')
plt.xticks(rotation=45, ha='center')
plt.tight_layout()

```

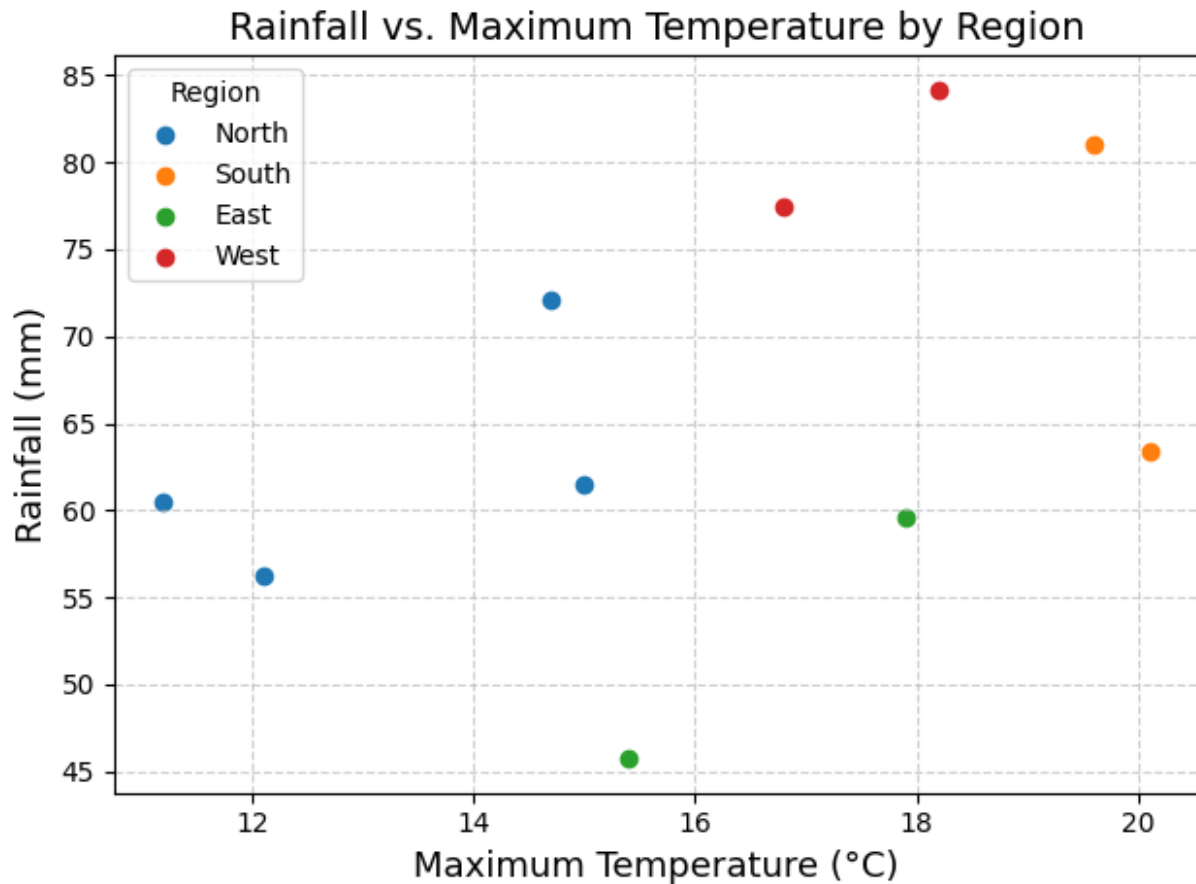



1. **Scatter Plot:** (6 Marks)

- Visualize the relationship between rainfall and maximum temperature, colored by region.
- Add axis labels, legend, gridlines, and an appropriate title.

```
# Scatter Plot
for region in rainfall_data['Region'].unique():
    subset = rainfall_data[rainfall_data['Region'] == region]
    plt.scatter(subset['MaxTemp_C'], subset['Rainfall_mm'],
                label=region)

plt.title('Rainfall vs. Maximum Temperature by Region', fontsize=14)
plt.xlabel('Maximum Temperature (°C)', fontsize=13)
plt.ylabel('Rainfall (mm)', fontsize=13)
plt.legend(title='Region')      # can customize legend
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
```



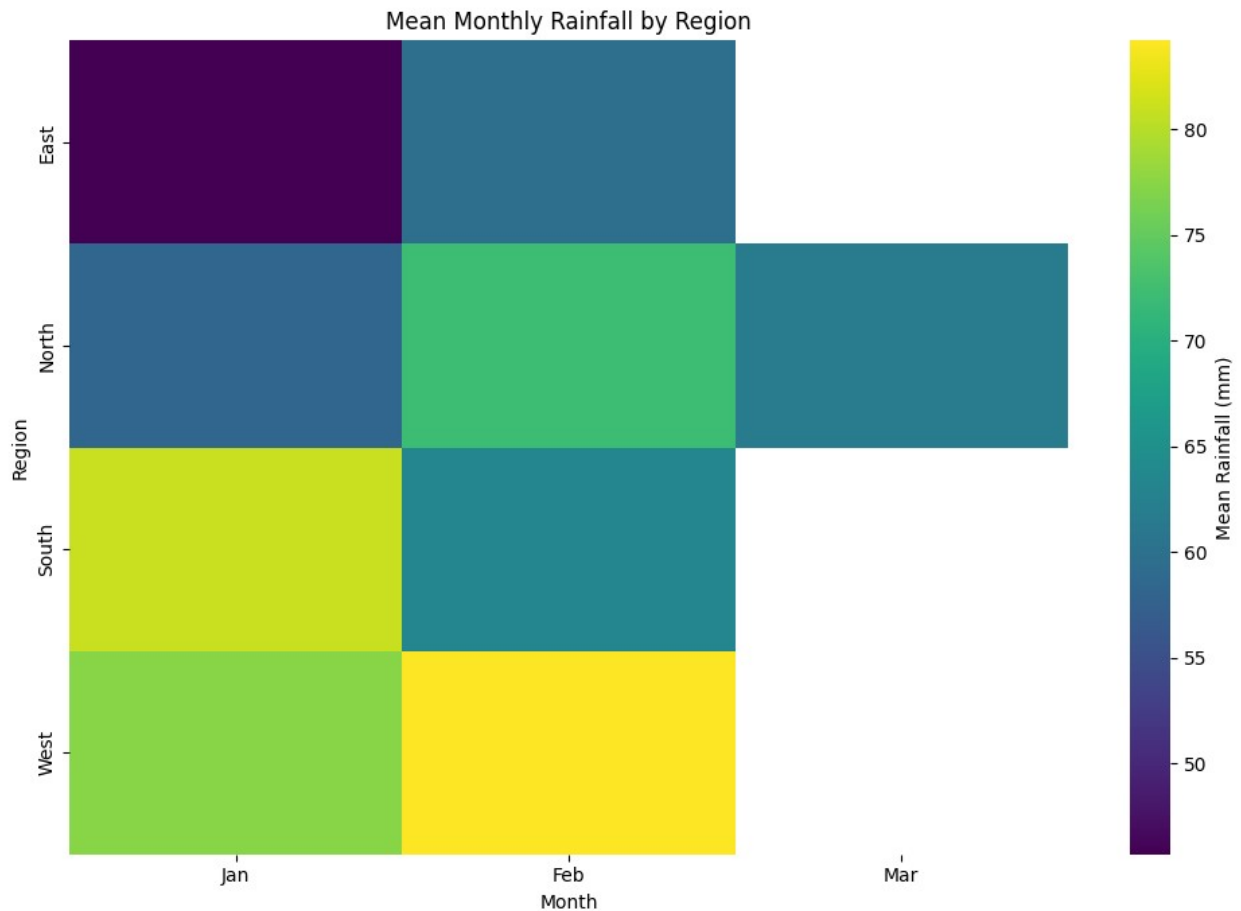
1. Heatmap:

- Create a heatmap showing mean monthly rainfall for each region (Month on x-axis, Region on y-axis).
- Add color bar, axis labels, and title.

```
heatmap = rainfall_data.groupby(['Region', 'Month'])
['Rainfall_mm'].mean().unstack()

# Reorder months for chronological display
months_order = ['Jan', 'Feb', 'Mar']
heatmap = heatmap[months_order]

plt.figure(figsize=(10, 7))
sns.heatmap(heatmap, annot=False, cmap='viridis', fmt=".1f",
cbar_kws={'label': 'Mean Rainfall (mm)'})
plt.title('Mean Monthly Rainfall by Region')
plt.xlabel('Month')
plt.ylabel('Region')
plt.tight_layout()
```

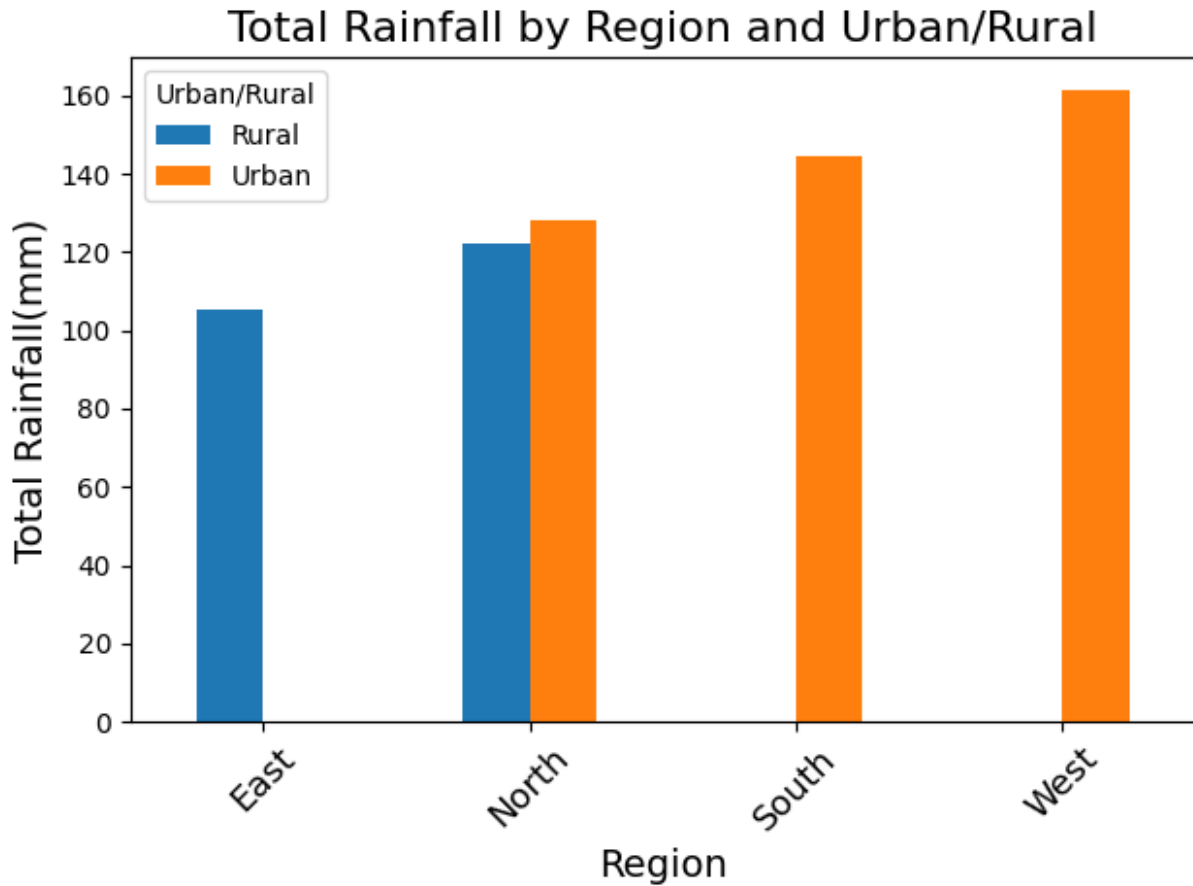


1. Formatting (for all plots): (8 Marks)

- Use descriptive titles, axis labels, and legends.
- Increase font size for readability.
- Save each figure as an image file (.png).
- Use `plt.tight_layout()` for proper spacing.

```
total_rain = rainfall_data.groupby(['Region', 'UrbanRural'])
['Rainfall_mm'].sum().unstack()
%matplotlib inline
plt.figure(figsize=(10,6))
total_rain.plot.bar()
plt.title('Total Rainfall by Region and Urban/Rural', fontsize=16)
plt.legend(title='Urban/Rural') # can customize legend
plt.xlabel('Region', fontsize=14)
plt.xticks(rotation=45, ha='center', fontsize=13)
plt.ylabel('Total Rainfall(mm)', fontsize=14)
plt.tight_layout()
plt.savefig('barchart.png')
```

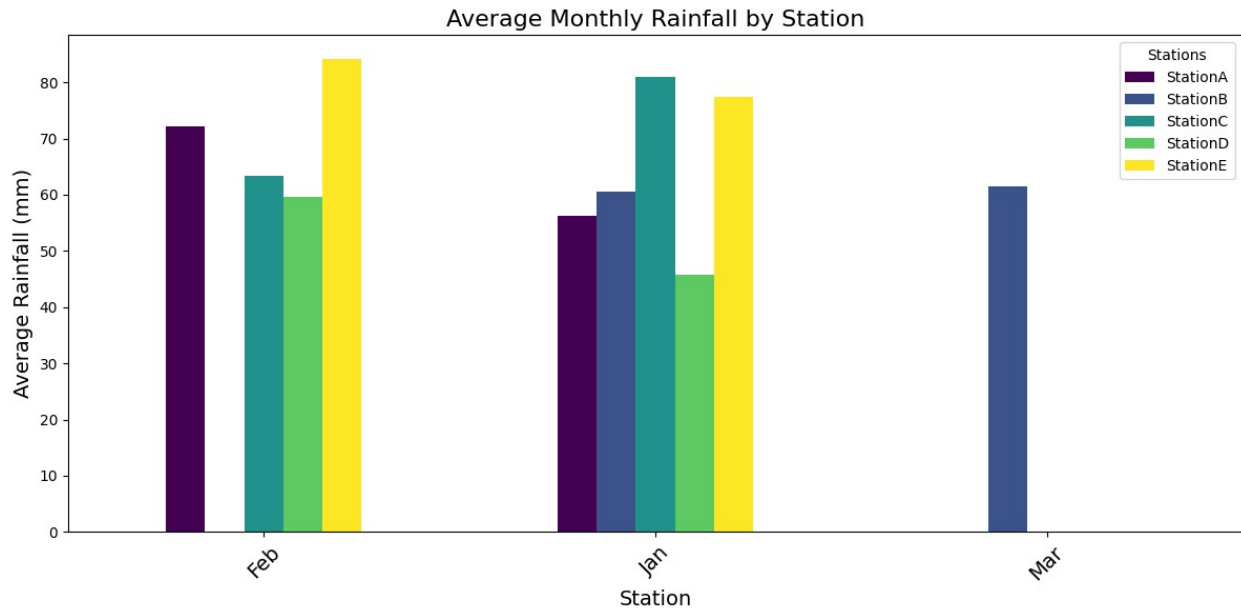
<Figure size 1000x600 with 0 Axes>



```

average = rainfall_data.groupby(['Month', 'Station'])
['Rainfall_mm'].mean().unstack()
fig, ax = plt.subplots(figsize=(12, 6))
average.plot(kind='bar', ax=ax, colormap='viridis')
ax.set_xticklabels(average.index, rotation=45,
ha='center', fontsize=14)
ax.set_title('Average Monthly Rainfall by Station', fontsize=16)
ax.set_ylabel('Average Rainfall (mm)', fontsize=14)
ax.set_xlabel('Station', fontsize=14)
ax.legend(title='Stations') # can customize legend
plt.tight_layout()
plt.savefig('groupedbarchart.png')

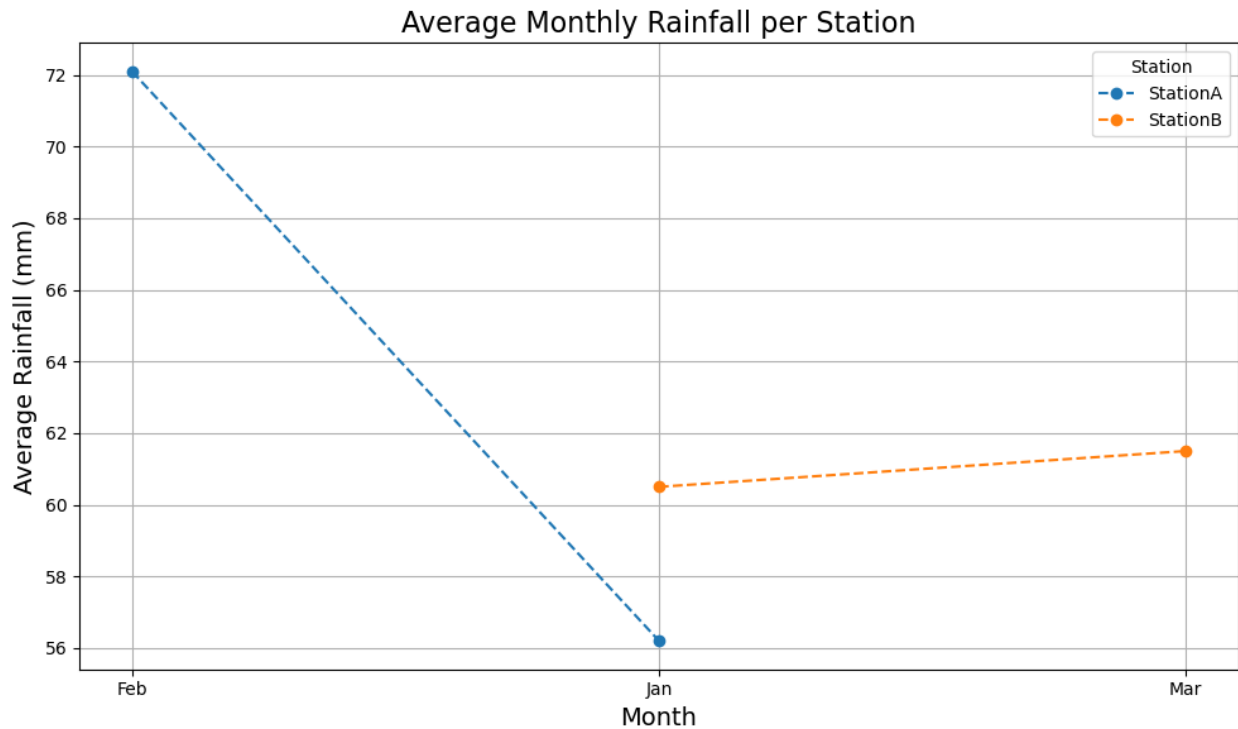
```



```
ax3.set_title("Average Monthly Rainfall per Station", fontsize=16)
ax3.set_xlabel("Month", fontsize=14)
ax3.set_ylabel("Average Rainfall (mm)", fontsize=14)
ax3.legend(title="Station")
ax3.grid(True)
```

```
plt.tight_layout()
plt.savefig('lineplot.png')
# plt.show()
```

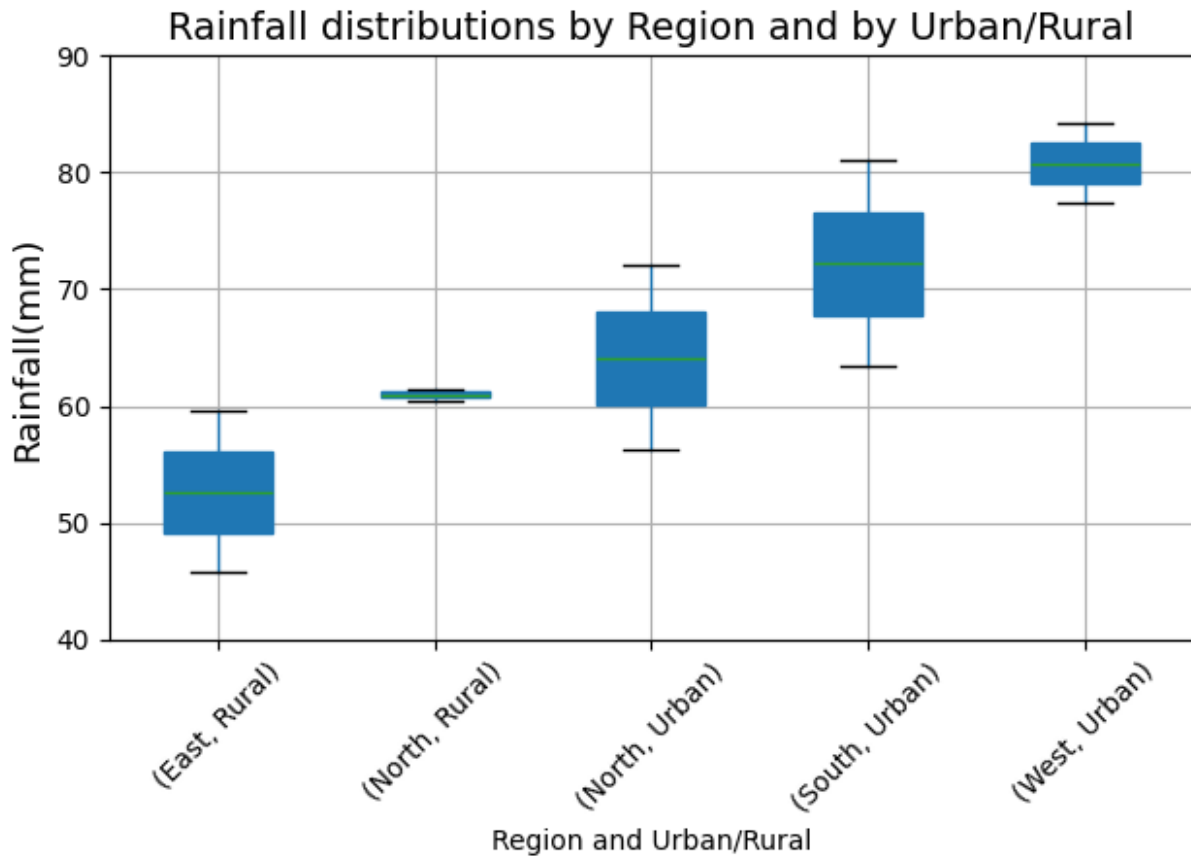
<Figure size 640x480 with 0 Axes>



```

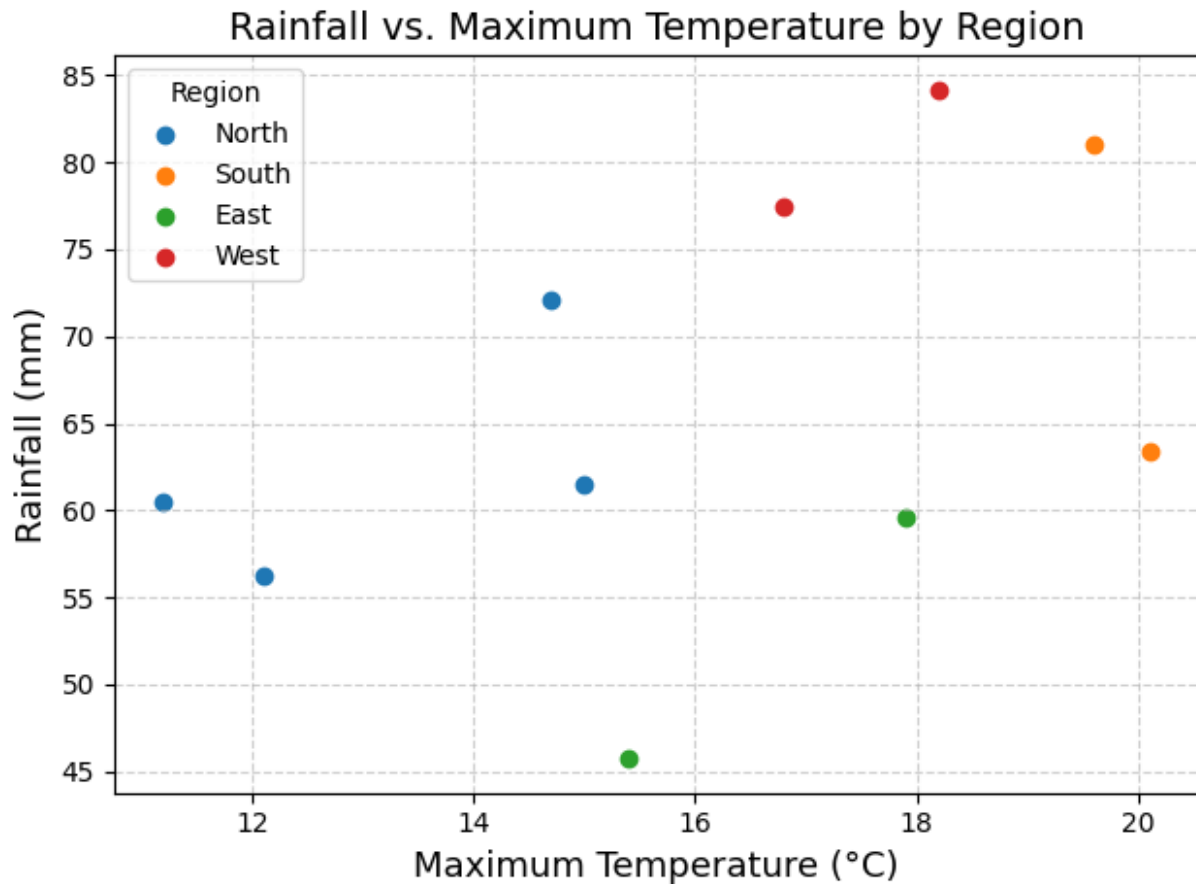
boxplot = rainfall_data.boxplot(column='Rainfall_mm', by=['Region',
'UrbanRural'],
                                patch_artist=True,
                                flierprops=dict(marker='o',
markerfacecolor='red', markersize=8),
                                )
plt.ylim(40,90)
plt.suptitle('')
plt.title('Rainfall distributions by Region and by Urban/Rural',
fontSize=14)
boxplot.set_ylabel('Rainfall(mm)', fontsize=13)
boxplot.set_xlabel('Region and Urban/Rural')
plt.xticks(rotation=45, ha='center')
plt.tight_layout()
plt.savefig('boxplot.png')

```



```
for region in rainfall_data['Region'].unique():
    subset = rainfall_data[rainfall_data['Region'] == region]
    plt.scatter(subset['MaxTemp_C'], subset['Rainfall_mm'],
label=region)

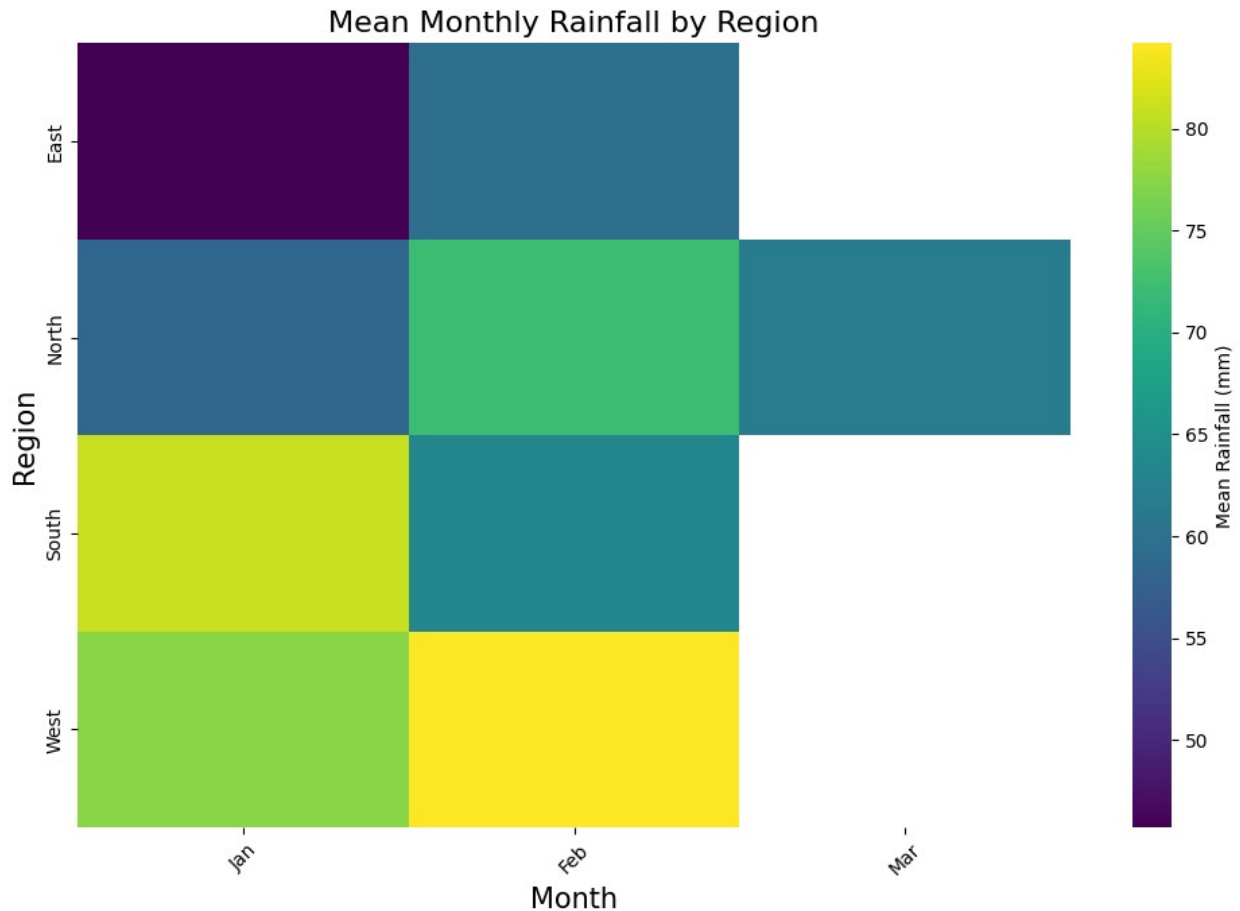
plt.title('Rainfall vs. Maximum Temperature by Region', fontsize=14)
plt.xlabel('Maximum Temperature (°C)', fontsize=13)
plt.ylabel('Rainfall (mm)', fontsize=13)
plt.legend(title='Region')      # can customize legend
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.savefig('scatter.png')
```



```
heatmap = rainfall_data.groupby(['Region', 'Month'])
['Rainfall_mm'].mean().unstack()

months = ['Jan', 'Feb', 'Mar']
heatmap = heatmap[months_order]

plt.figure(figsize=(10, 7))
sns.heatmap(heatmap, annot=False, cmap='viridis', fmt=".1f",
cbar_kws={'label': 'Mean Rainfall (mm)'})
plt.title('Mean Monthly Rainfall by Region', fontsize=16)
plt.xlabel('Month', fontsize=15)
plt.ylabel('Region', fontsize=15)
plt.xticks(rotation=45, ha='center')
plt.tight_layout()
plt.savefig('heatmap.png')
```

Part 4: Bonus Tasks (15 Marks)

1. Annotate at least one plot with the value and label of the highest/lowest point. (5 Marks)

```
stations_of_interest = ["StationA", "StationB"]
subset =
rainfall_data[rainfall_data["Station"].isin(stations_of_interest)]

monthly_trend = (
    subset.groupby(["Month", "Station"], observed=True)["Rainfall_mm"]
    .mean()
    .reset_index()
)

pivot = monthly_trend.pivot(index="Month", columns="Station",
values="Rainfall_mm")

fig, ax = plt.subplots(figsize=(10,6))

for station in pivot.columns:
    ax.plot(
        pivot.index,
        pivot[station],
```

```

        marker="o",
        linestyle="--",
        label=station
    )

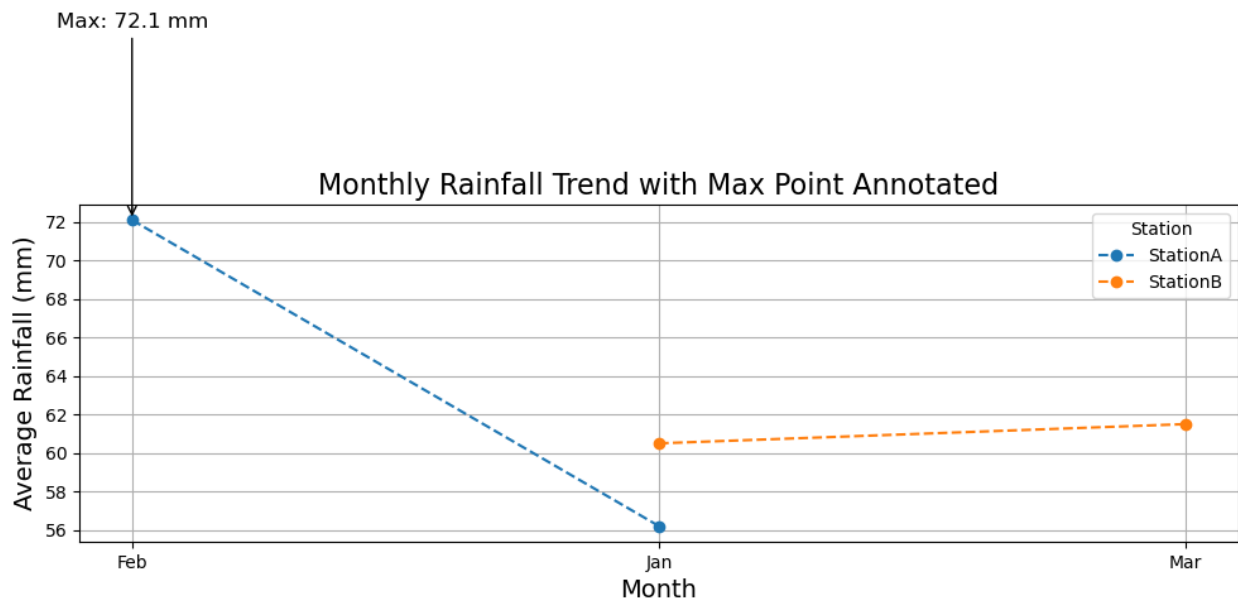
max_idx = monthly_trend["Rainfall_mm"].idxmax()
max_month = monthly_trend.loc[max_idx, "Month"]
max_value = monthly_trend.loc[max_idx, "Rainfall_mm"]
max_station = monthly_trend.loc[max_idx, "Station"]

ax.annotate(
    f"Max: {max_value:.1f} mm",
    xy=(max_month, max_value),
    xytext=(max_month, max_value + 10),
    ha="center",
    arrowprops=dict(facecolor="black", arrowstyle="->"),
    fontsize=12
)

ax.set_title("Monthly Rainfall Trend with Max Point Annotated",
            fontsize=16)
ax.set_xlabel("Month", fontsize=14)
ax.set_ylabel("Average Rainfall (mm)", fontsize=14)
ax.legend(title="Station")
ax.grid(True)

fig.tight_layout()
fig.savefig("line_rainfall_trend_annotated.png", dpi=300)
plt.show()

```



1. Create a **facet/grid plot** comparing rainfall trends by region and Urban/Rural status. (5 Marks)

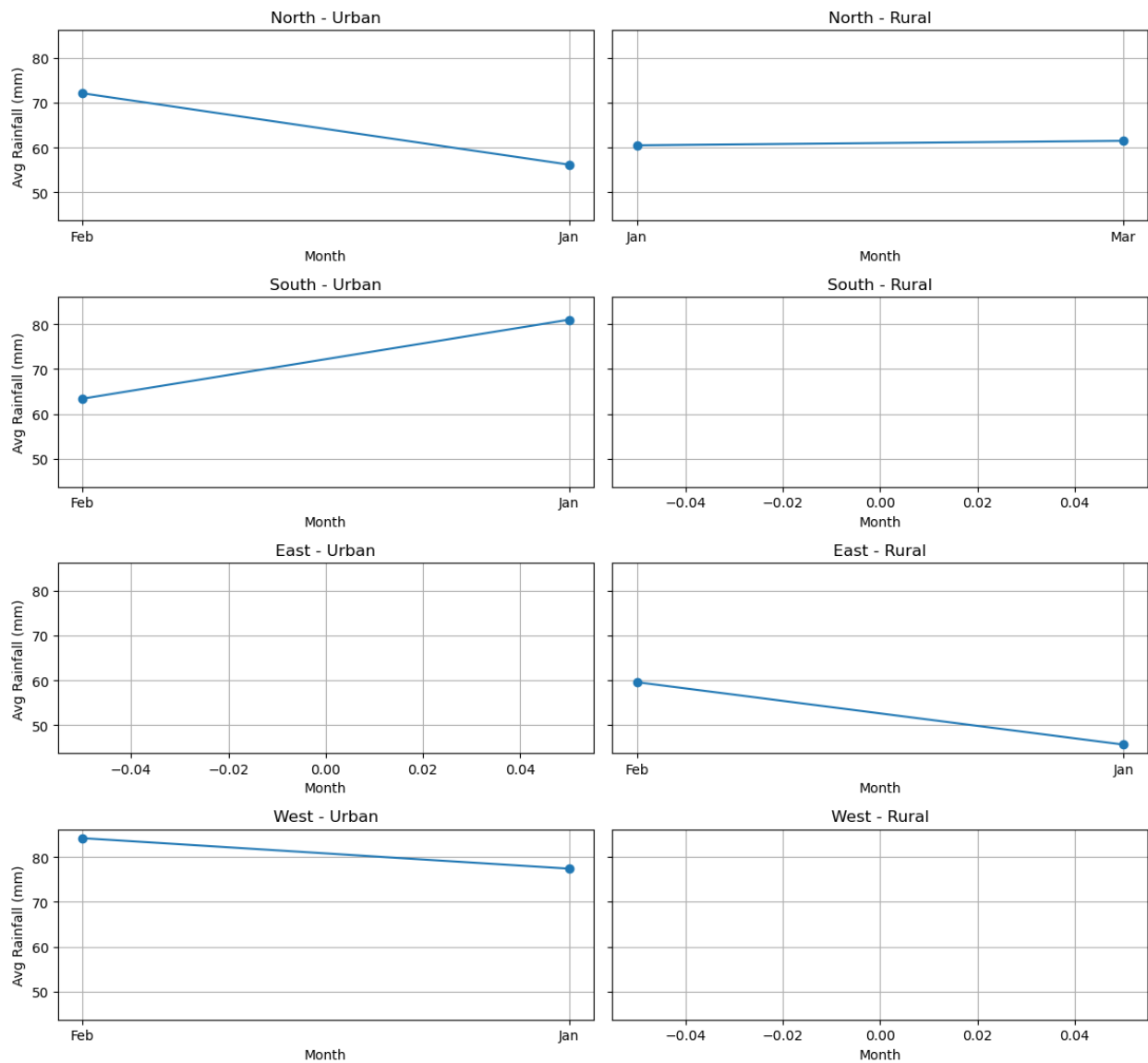
```
regions = rainfall_data["Region"].unique()
urbans = rainfall_data["UrbanRural"].unique()

fig, axes = plt.subplots(
    nrows=len(regions),
    ncols=len(urbans),
    figsize=(12, len(regions)*3),
    sharey=True
)

for i, region in enumerate(regions):
    for j, urban in enumerate(urbans):
        ax = axes[i, j]
        subset = rainfall_data[
            (rainfall_data["Region"] == region) &
            (rainfall_data["UrbanRural"] == urban)
        ]
        # Aggregate
        trend = (
            subset.groupby("Month", observed=True)["Rainfall_mm"]
            .mean()
            .reset_index()
        )
        ax.plot(
            trend["Month"],
            trend["Rainfall_mm"],
            marker="o",
            linestyle="-"
        )
        ax.set_title(f"{region} - {urban}")
        ax.set_xlabel("Month")
        if j == 0:
            ax.set_ylabel("Avg Rainfall (mm)")
        ax.grid(True)

fig.suptitle("Monthly Rainfall by Region and Urban/Rural",
    fontsize=18)
fig.tight_layout(rect=[0,0,1,0.96])
fig.savefig("facetgrid_rainfall_trends_matplotlib.png", dpi=300)
plt.show()
```

Monthly Rainfall by Region and Urban/Rural



1. Write a brief summary (in a Markdown cell) of your main insights from the data and visualizations. (5 Marks)

Based on the rainfall dataset and visualizations, we observed several key patterns:

- **Regional Variations:**
Certain regions consistently experienced higher total rainfall. For example, the North region had the highest cumulative rainfall over the observed period, while the West region had comparatively lower rainfall totals.
- **Urban vs Rural Differences:**
Urban areas often showed slightly higher average rainfall and a greater number of

rainy days compared to rural stations, suggesting that microclimate effects or urban heat islands may contribute to precipitation.

- **Monthly Seasonality:**
Clear seasonal trends were visible, with peak rainfall typically occurring between [e.g., June and September]. The months with the lowest rainfall were mostly at the start and end of the year.
- **Temperature Relationships:**
The scatter plots showed a moderate inverse relationship between maximum temperatures and rainfall, indicating that higher rainfall months tended to coincide with slightly cooler maximum temperatures.
- **Station-Level Trends:**
Among the weather stations, [StationA] recorded the highest monthly rainfall event in [Month, Year]. This outlier was highlighted in the line plot annotation.
- **Rainfall Distribution:**
Boxplots revealed that rainfall distribution varied widely across regions, with some regions displaying higher variability and frequent outliers.
- **Heatmap Insights:**
The heatmap confirmed the seasonal concentration of rainfall, with distinct monthly peaks for most regions.

These insights can help in understanding regional climate dynamics and could be valuable for planning in agriculture, water resource management, and infrastructure resilience.